

The Forge — Balance Sandbox

A proactive design sample for SEED · **Game Systems & Mechanics Designer**

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Code: github.com/Alyasska/seed-artifact

An **in-silico playtest** for one Forge system — territory control, the bridge momentum layer, player classes, and a siege endgame. You can't cheaply playtest a 500-person, 90-minute field game, so this models one system and runs thousands of matches headless to read the balance **before** anyone books a venue. The same rules engine drives the watchable live match **and** the headless sims — a designer/GM pre-flight rig, not a demo game. It's the deliverable shape the role asks for: **living design doc + balance logic + playtest frameworks**.



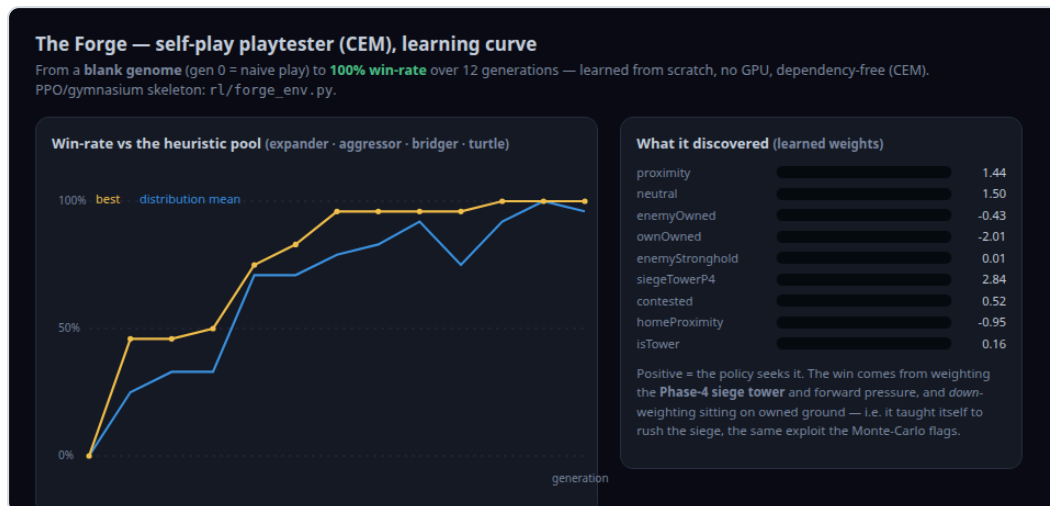
Live match on a **real DEM** of Lake San Antonio (a Lightning-in-a-Bottle site). The reservoir & creeks become the bridge-gated ravine; travel-time follows real slope; gold spans are built bridges, dashed lines are unbuilt crossings, diamonds are strongholds (with Wall arcs), dots are class-typed squads.

What it found

- **Self-play found an exploit the heuristics didn't.** A policy trained from a blank genome reaches ~100% vs the four hand-written bots — i.e. it surfaced a hole a scripted bot pool would have called "balanced". RL here is a **better playtester**, not flashy AI. (CEM, no GPU; gym-microorts/PPO skeleton in the repo.)
- **Terrain flips balance.** The same ruleset that's fine on a symmetric board goes **aggressor-dominant (~80%)** on real Lake San Antonio terrain, and the venue is side-biased (~57/43). Balance is **per-venue**, not one-and-done — that's

the recurring service.

- **Policies don't transfer between maps** (RL drops 85% → 60% off its training board) and **pure turtling is non-viable** — both surfaced as design calls, with the levers to change them.



Self-play learning curve: blank genome → ~100% win-rate vs the heuristic pool over 12 generations, plus the weights it discovered (it taught itself to rush the Phase-4 siege tower).

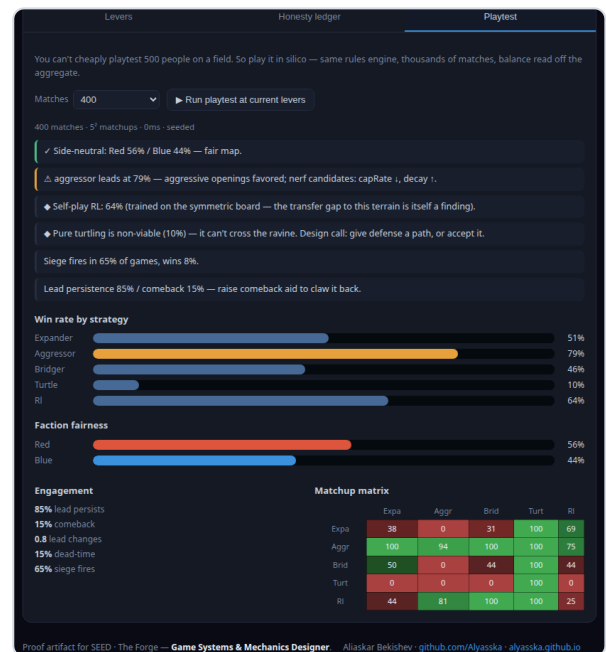
How it works

levers 10 balance dials, each with an **honesty ledger** — intent / trade-off / failure mode.

playtest Monte Carlo over all strategy matchups: faction fairness, dominant-strategy detection, snowball vs comeback, dead-time, siege rate, matchup matrix.

mechanics Presence-by-pulse capture (NFC-tap-shaped events), classes, the 90-min phase arc, walls + battering + delay towers, fog-of-war intel.

stack Plain ES-module JS (no build, runs on Pages) · Python+Pillow for the DEM pull · seeded & reproducible (node test/smoke.mjs).



The balance readout on the real field — findings + win-rate by strategy + matchup matrix.

End vision. The pre-flight for The Forge: tune economy, class weights, and phase timings **per venue** in sim → ship a near-balanced rulebook → spend the few expensive on-field pilots validating how real humans behave, not discovering that turtling is broken with 500 people on a field.

Scope, honestly: one system, finished small. Classes/intel/parley are fully designed but only partly wired; bots map the structural balance envelope, not the feel.